C1: Search Extensions

FIT1061 Credit Task — Week 5 — When Algorithms Choose Who Gets Help

Credit tasks are designed to push your understanding beyond the core Pass-level implementation. This task asks you to explore how search methods can be extended, compared, and adapted to new conditions rather than simply applying the same procedure in a familiar setting.

Show us that you can

- You can move beyond the baseline implementation and reason about extensions to the search strategy.
- You can explain how your extension changes behaviour or performance.
- You can connect the extension back to the earlier search tasks and their tradeoffs.

Give us

- A completed notebook with an implementation showing the extension.
- A short explanation of what changed and why it matters.
- Any required reflection or comparison response.

Resources

- Week 4–5 lecture material *(Moodle)*
- C1 starter notebook *(OnTrack)*
- Relevant search notes from earlier weeks *(Moodle)*

Key sections in the task notebook

- C1.1: Random Restarts: Making Hill Climbing Reliable
- C1.2: Multi-Graph Comparison: How Structure Shapes Search
- Check Your Understanding

Do the following

1. Read the task instructions and review the earlier search methods (random restarts is introduced in the Week 4 lecture; the graph-structure comparison builds on Week 5).
2. Implement and test the extension carefully.
3. Write an explanation of the change in your own words.
4. Submit the completed work and mark it ready for feedback.

! Acceptable use of Artificial Intelligence

AI tools are **discouraged** for the core analytical work at the Credit level. You may use AI for *explanation* (e.g., “explain the search extension”), but not to generate the core analysis or write your reflection. Cite all AI use. The milestone discussion will probe for genuine understanding.
All submitted work must be your own.